



Computer Network

Network Address Translation
(NAT)



IPv4 Private Address space



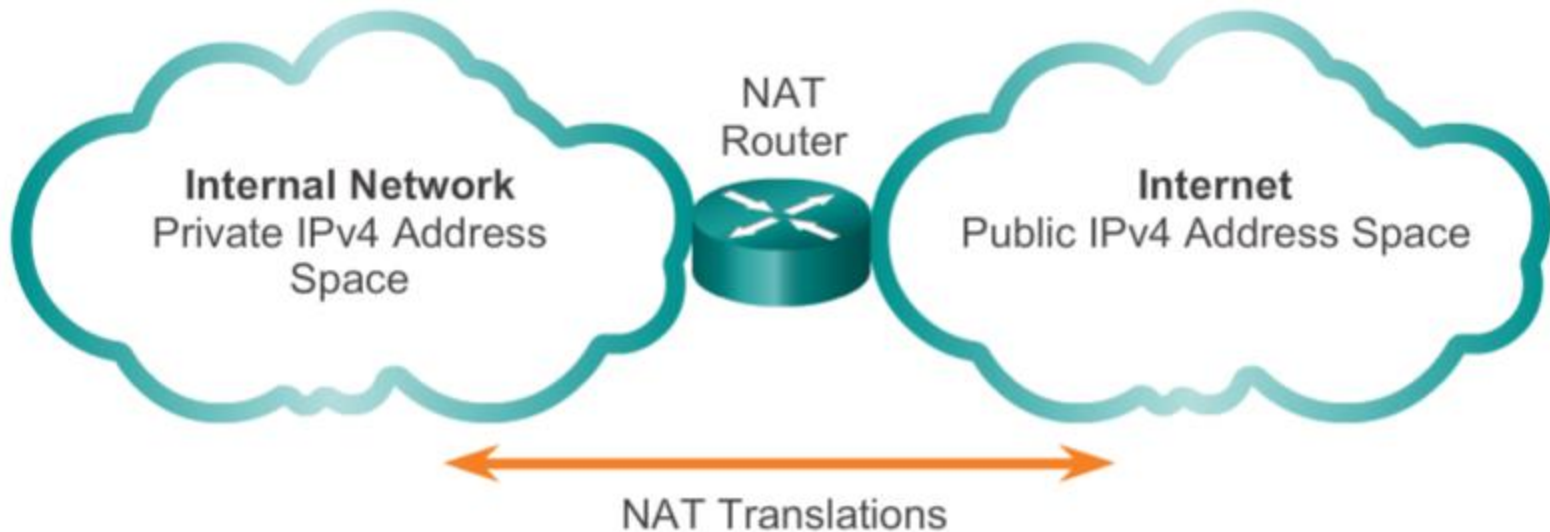
Private Internet addresses are defined in RFC 1918:

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16



Address space

- The Internet is run out of address space
- Solution
 - Network Address Translation (NAT)



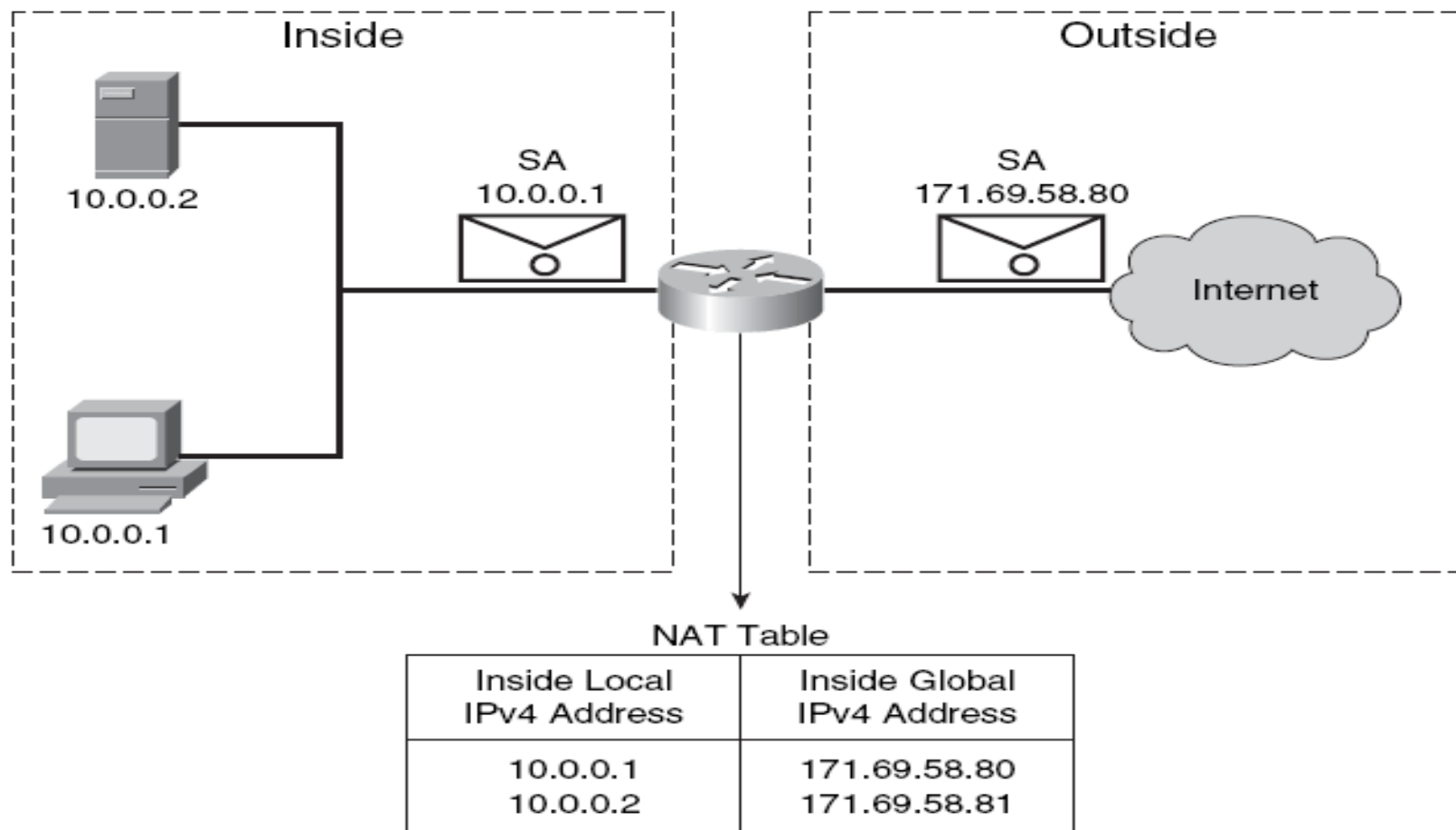


NAT introduction

- IP private & IP public
- NAT translates private address to public address for outcomming traffic and vice versa
- NAT allow private network to connect to the Internet



NAT introduction





NAT category

- Static NAT
 - Map an IP private to an IP public (one to one)
- Dynamic NAT
 - Map multiple IP private to an IP public from a group of IP public
- NAT overloading (PAT)
 - Map multiple IP private to a single IP public

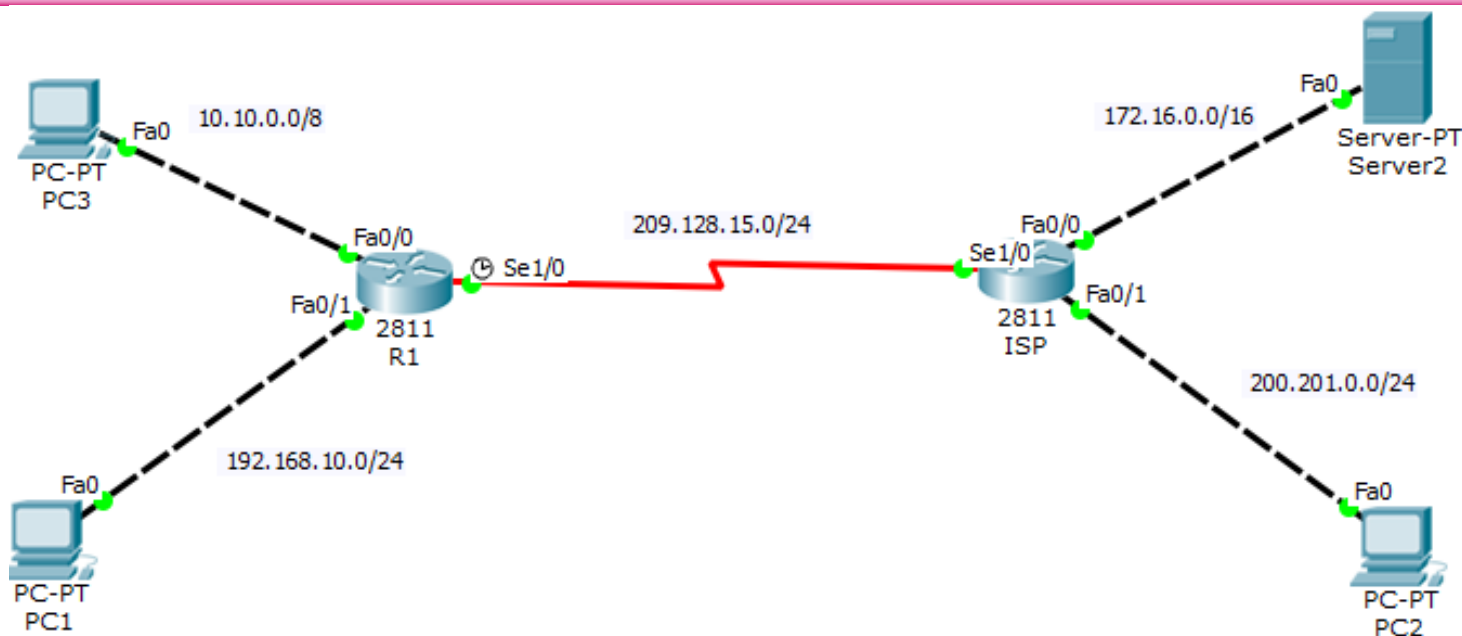


Static NAT Configure

- Step 1
 - RouterX(config)# **ip nat inside source static** *local-ip* *global-ip*
- Step 2 (inside interface)
 - RouterX(config-if)# **ip nat inside**
- Step 3 (outside interface)
 - RouterX(config-if)# **ip nat outside**



Static NAT Configure



```
R1(config)#ip nat inside source static 10.10.0.2 209.128.15.3
R1(config)#interface se1/0
R1(config-if)#ip nat outside
R1(config-if)#interface fa0/0
R1(config-if)#ip nat inside
```



Static NAT Verification



- Show NAT table in router

```
R1#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.128.15.3:17	10.10.0.2:17	172.16.0.2:17	172.16.0.2:17

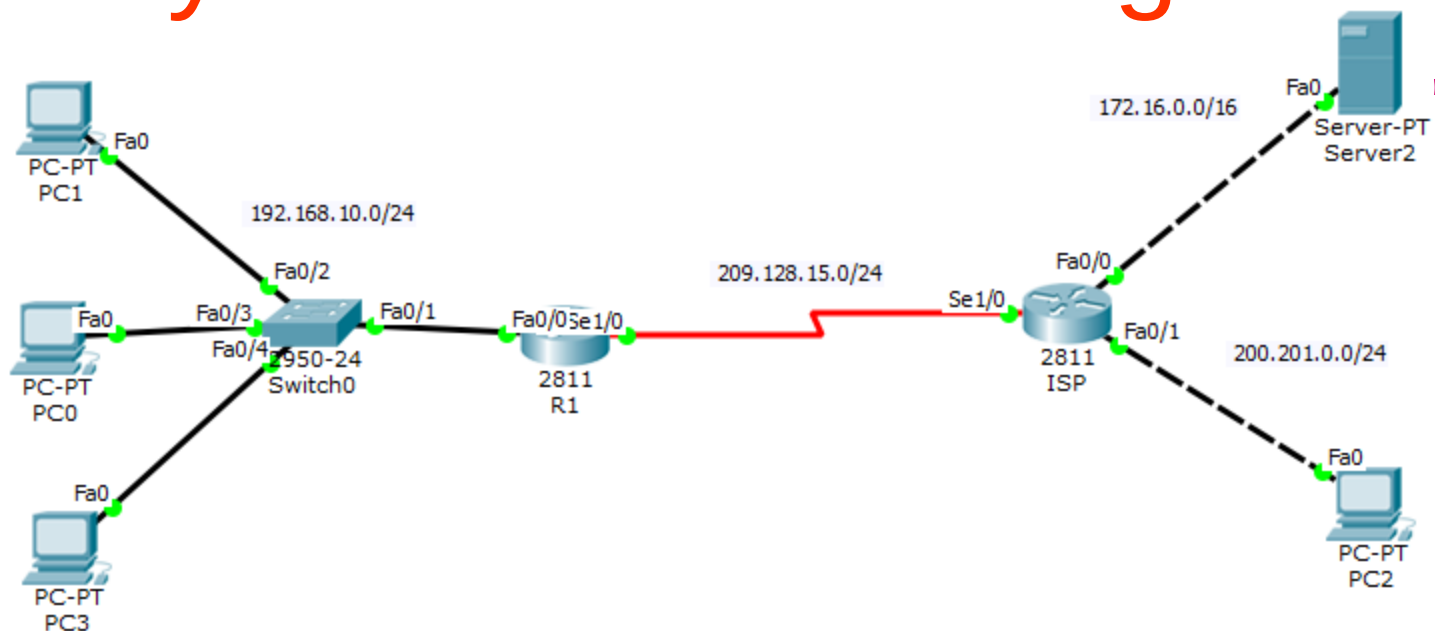


Dynamic NAT configure

- Step 1: define a pool of global addresses
 - RouterX(config)# **ip nat pool** *name start-ip end-ip {netmask netmask |prefix-length prefix-length}*
- Step 2: define a list of private network
 - RouterX(config)# **access-list** *access-list-number permit source [sourcewildcard]*
- Step 3: enable dynamic NAT
 - RouterX(config)# **ip nat inside source list** *access-list-number pool name*
- Step 4: inside interface
 - RouterX(config-if)# **ip nat inside**
- Step 5: outside interface
 - RouterX(config-if)# **ip nat outside**



Dynamic NAT Configure



```
R1(config)#ip nat pool MyPool 209.128.15.3 209.128.15.10 netmask 255.255.255.0
```

```
R1(config)#access-list 15 permit 192.168.10.0 0.0.0.255
```

```
R1(config)#ip nat inside source list 15 pool MyPool
```

```
R1(config)#interface se1/0
```

```
R1(config-if)#ip nat outside
```

```
R1(config-if)#interface fa0/0
```

```
R1(config-if)#ip nat inside
```



Dynamic NAT Verification



- Show NAT table in router

```
R1#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.128.15.5:4	192.168.10.2:4	172.16.0.2:4	172.16.0.2:4
icmp	209.128.15.6:3	192.168.10.3:3	172.16.0.2:3	172.16.0.2:3
icmp	209.128.15.7:2	192.168.10.4:2	172.16.0.2:2	172.16.0.2:2



NAT benefit

- Save address space
- Protect network security



NAT overloading (PAT)

- (Port Address Translation)
- TCP/UDP
- Port number
- PAT allows multi hosts connect Internet via 1 public IP address

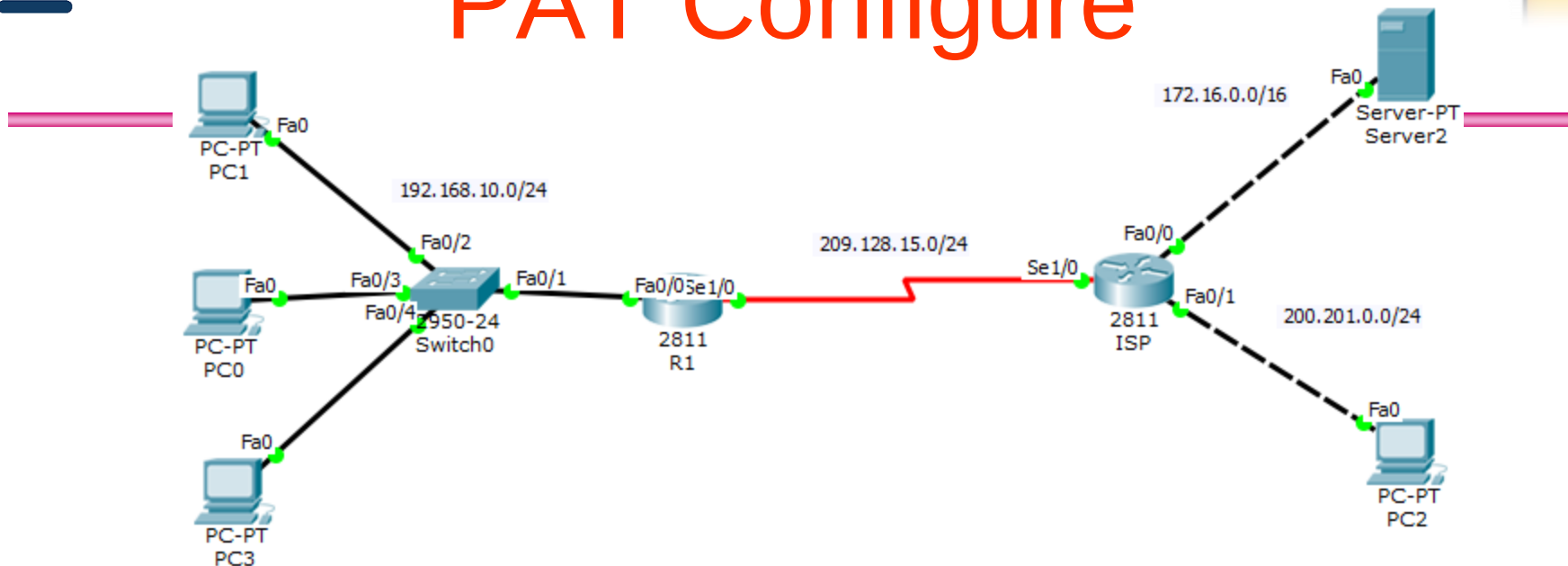


PAT configuration

- Step 1
 - Router(config)# **access-list** *access-list-number* **permit** *source* [*source-wildcard*]
- Step 2
 - Router(config)#**ip nat inside source list** *access-list-number* **interface** *interface* **overload**
- Step 3
 - Router(config-if)#ip nat outside
 - Router(config-if)#ip nat outside



PAT Configure



```
Router(config)#access-list 15 permit 192.168.10.0 0.0.0.255
Router(config)#ip nat inside source list 15 interface se1/0 overload
Router(config)#interface se1/0
Router(config-if)#ip nat outside
Router(config-if)#interface fa0/0
Router(config-if)#ip nat inside
```



PAT Verification

```
Router#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.128.15.2:1024	192.168.10.4:1	172.16.0.2:1	172.16.0.2:1024
icmp	209.128.15.2:1	192.168.10.3:1	172.16.0.2:1	172.16.0.2:1
icmp	209.128.15.2:3	192.168.10.2:3	172.16.0.2:3	172.16.0.2:3



NAT troubleshooting

- Show ip nat translation
- Show ip nat statistics
- Debug ip nat detailed